Proof and Generalization of a Trigonometric Product Identity

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Problem Statement

Prove the identity

$$\cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cos\frac{\alpha}{8} = \frac{\sin\alpha}{8\sin\frac{\alpha}{8}} \tag{1}$$

and generalize the result.

Proof of the Identity

We begin with the left-hand side (LHS) of the identity:

$$\text{LHS} = \cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cos\frac{\alpha}{8}$$

To evaluate this product, we multiply and divide the expression by $8\sin\frac{\alpha}{8}$:

$$\text{LHS} = \frac{8\sin\frac{\alpha}{8}\cos\frac{\alpha}{8}\cos\frac{\alpha}{4}\cos\frac{\alpha}{2}}{8\sin\frac{\alpha}{8}}$$

Recall the double-angle identity for sine: $\sin(2\theta) = 2\sin\theta\cos\theta$. We apply this identity repeatedly to the numerator to collapse the terms.

First, with $\theta = \frac{\alpha}{8}$, we have $2\sin\frac{\alpha}{8}\cos\frac{\alpha}{8} = \sin\frac{\alpha}{4}$. Substituting this back gives:

$$\text{LHS} = \frac{4\left(2\sin\frac{\alpha}{8}\cos\frac{\alpha}{8}\right)\cos\frac{\alpha}{4}\cos\frac{\alpha}{2}}{8\sin\frac{\alpha}{8}} = \frac{4\sin\frac{\alpha}{4}\cos\frac{\alpha}{4}\cos\frac{\alpha}{2}}{8\sin\frac{\alpha}{8}}$$

Next, apply the identity again with $\theta = \frac{\alpha}{4}$ to get $2\sin\frac{\alpha}{4}\cos\frac{\alpha}{4} = \sin\frac{\alpha}{2}$:

$$\text{LHS} = \frac{2\left(2\sin\frac{\alpha}{4}\cos\frac{\alpha}{4}\right)\cos\frac{\alpha}{2}}{8\sin\frac{\alpha}{8}} = \frac{2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}}{8\sin\frac{\alpha}{8}}$$

Finally, apply the identity one more time with $\theta = \frac{\alpha}{2}$ to get $2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2} = \sin\alpha$:

$$\text{LHS} = \frac{\sin\alpha}{8\sin\frac{\alpha}{8}}$$

This matches the right-hand side, completing the proof of the specific identity.

Generalization

The given identity is a specific case of a more general product of cosines where the angle is repeatedly halved. The general form for n terms is:

$$\prod_{k=1}^{n}\cos\frac{\alpha}{2^k} = \cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cdots\cos\frac{\alpha}{2^n} = \frac{\sin\alpha}{2^n\sin\frac{\alpha}{2^n}} \tag{2}$$

Proof. Let P_n be the product of the first n terms:

$$P_n = \cos\frac{\alpha}{2}\cos\frac{\alpha}{2^2}\cdots\cos\frac{\alpha}{2^n}$$

Multiply both sides by $2^n\sin\frac{\alpha}{2^n}$:

$$2^n\sin\frac{\alpha}{2^n}P_n = 2^{n-1}\left(2\sin\frac{\alpha}{2^n}\cos\frac{\alpha}{2^n}\right)\cos\frac{\alpha}{2^{n-1}}\cdots\cos\frac{\alpha}{2}$$

Using the double-angle identity $2\sin\theta\cos\theta = \sin(2\theta)$, the term in parentheses simplifies to $\sin\frac{\alpha}{2^{n-1}}$. The expression becomes:

$$2^n\sin\frac{\alpha}{2^n}P_n = 2^{n-1}\sin\frac{\alpha}{2^{n-1}}\cos\frac{\alpha}{2^{n-1}}\cdots\cos\frac{\alpha}{2}$$

Repeating this process of collapsing the sine and cosine terms $n-1$ more times creates a telescoping effect:

$$2^n\sin\frac{\alpha}{2^n}P_n = 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2} = \sin\alpha$$

Dividing by $2^n \sin \frac{\alpha}{2^n}$ (assuming $\sin \frac{\alpha}{2^n} \neq 0$), we arrive at the generalized formula:

$$P_n = \frac{\sin \alpha}{2^n \sin \frac{\alpha}{2^n}}$$

$\square$